Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.3631146448360478, median 0.288437662290549, std: 0.3108870803894683
Gyroscope error (imu0): mean 0.5272086202552259, median 0.2611880923692109, std: 1.082511361831832
Accelerometer error (imu0): mean 0.6200678922828137, median 0.3867448558445936, std: 1.1979898745911388

Residuals

Reprojection error (cam0) [px]: mean 0.3631146448360478, median 0.288437662290549, std: 0.3108870803894683
Gyroscope error (imu0) [rad/s]: mean 0.02081638210668947, median 0.010312788758732186, std: 0.04274203659988589
Accelerometer error (imu0) [m/s^2]: mean 0.2407400726573361, median 0.15015288786056535, std: 0.46511708321172385

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.04106544 -0.99199547 0.11940942 0.00270167]
 [0.17629639 -0.12482904 -0.97638993 0.02919306]
 [0.98348015 -0.01904443 0.18001138 -0.0328033]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.04106544 0.17629639 0.98348015 0.02722571]
 [-0.99199547 -0.12482904 -0.01904443 0.00569947]
 [0.11940942 -0.97638993 0.18001138 0.03408617]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.004108977340000053

Gravity vector in target coords: [m/s^2]
[0.18783813 -9.80063407 -0.28409772]

Calibration configuration
==========

cam0

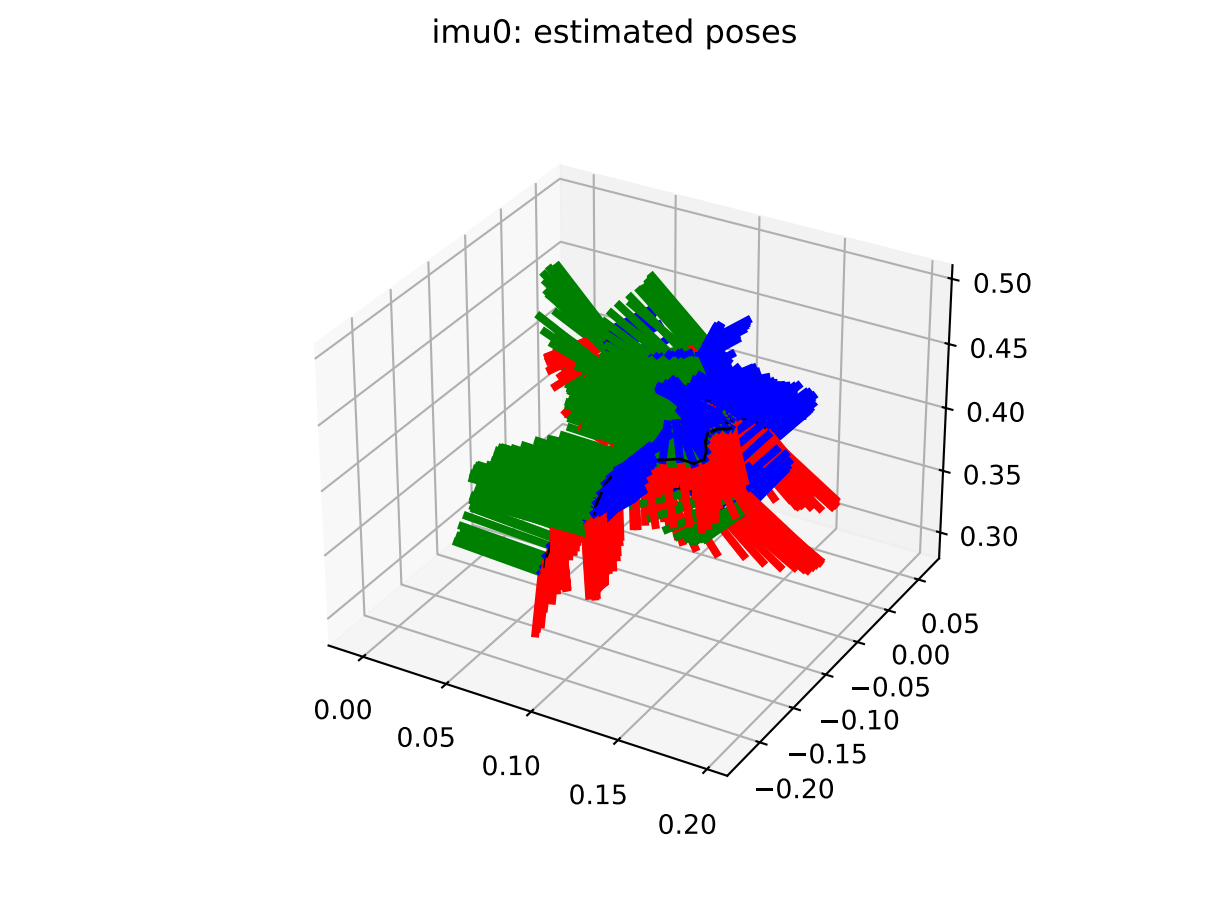
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.02 [m]
    Spacing 0.006 [m]

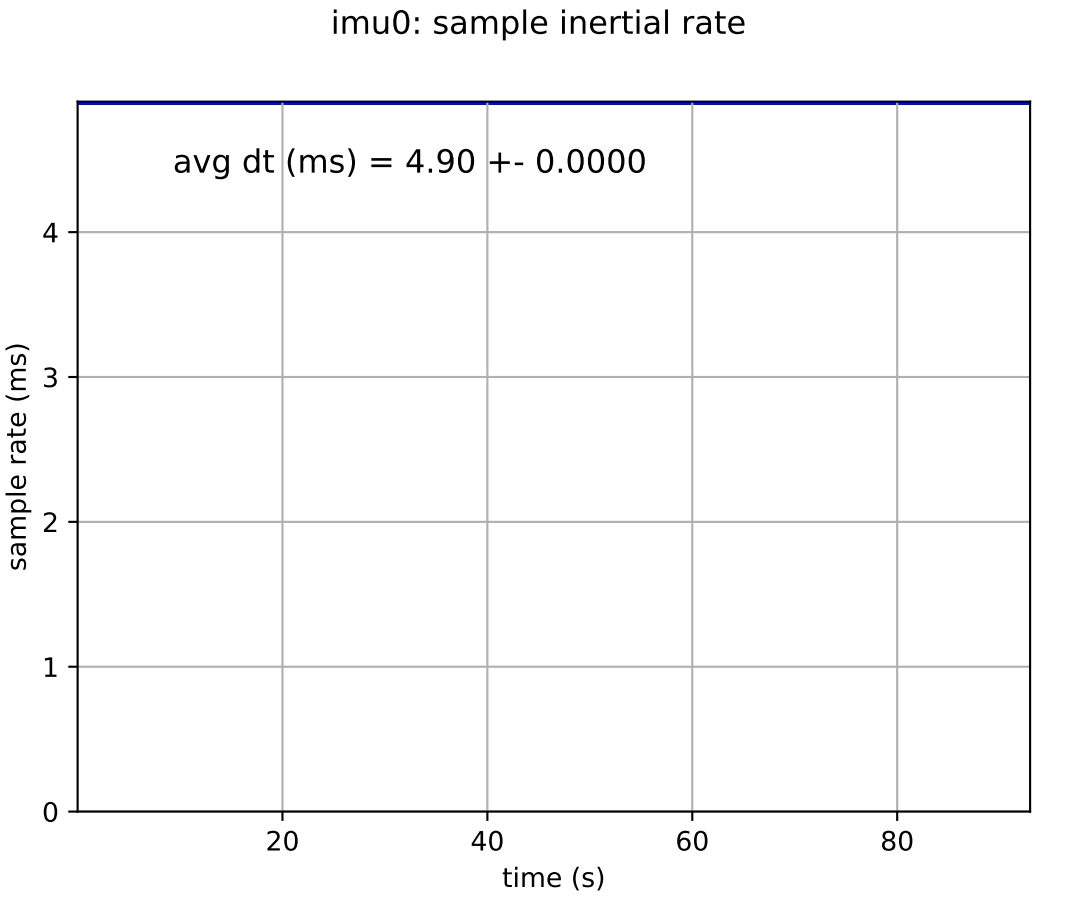
IMU configuration
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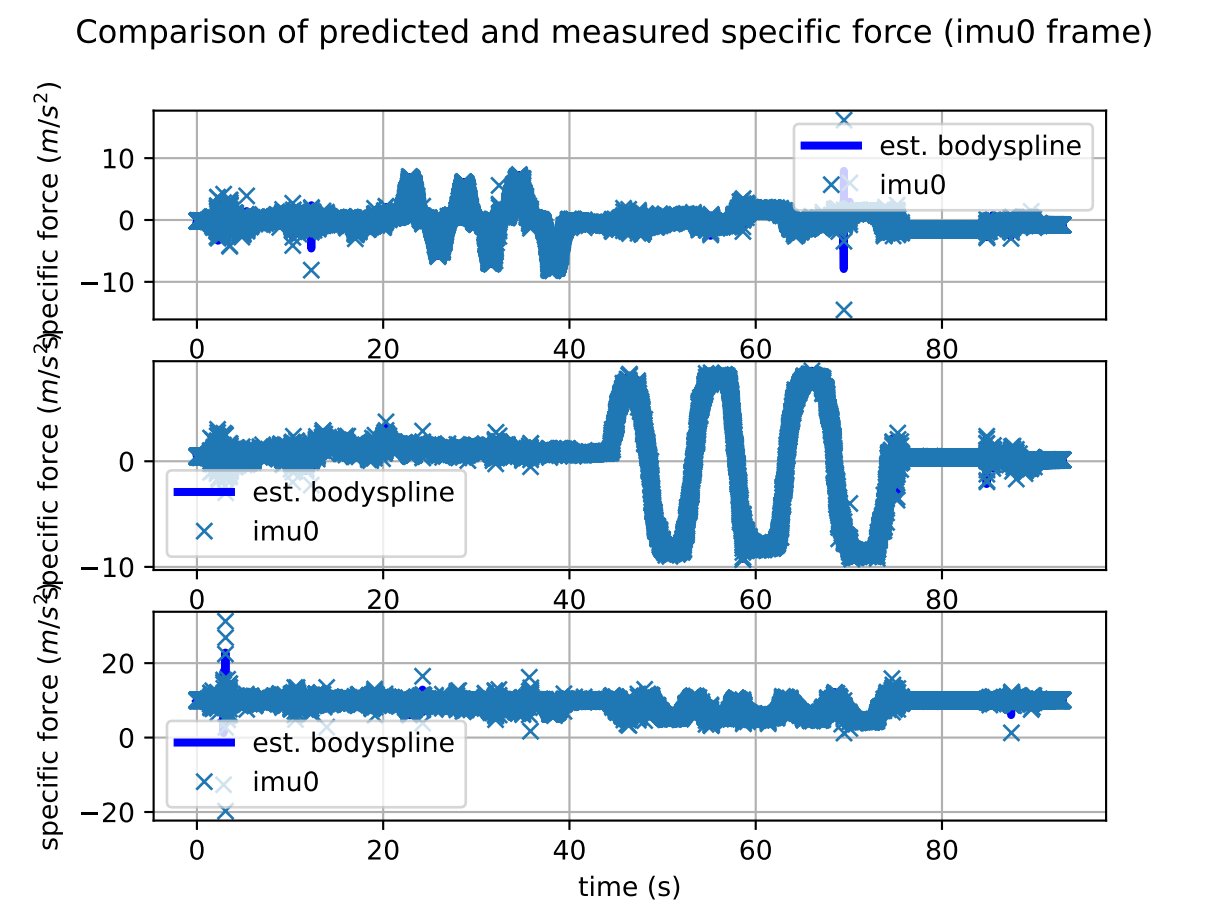
IMU0:

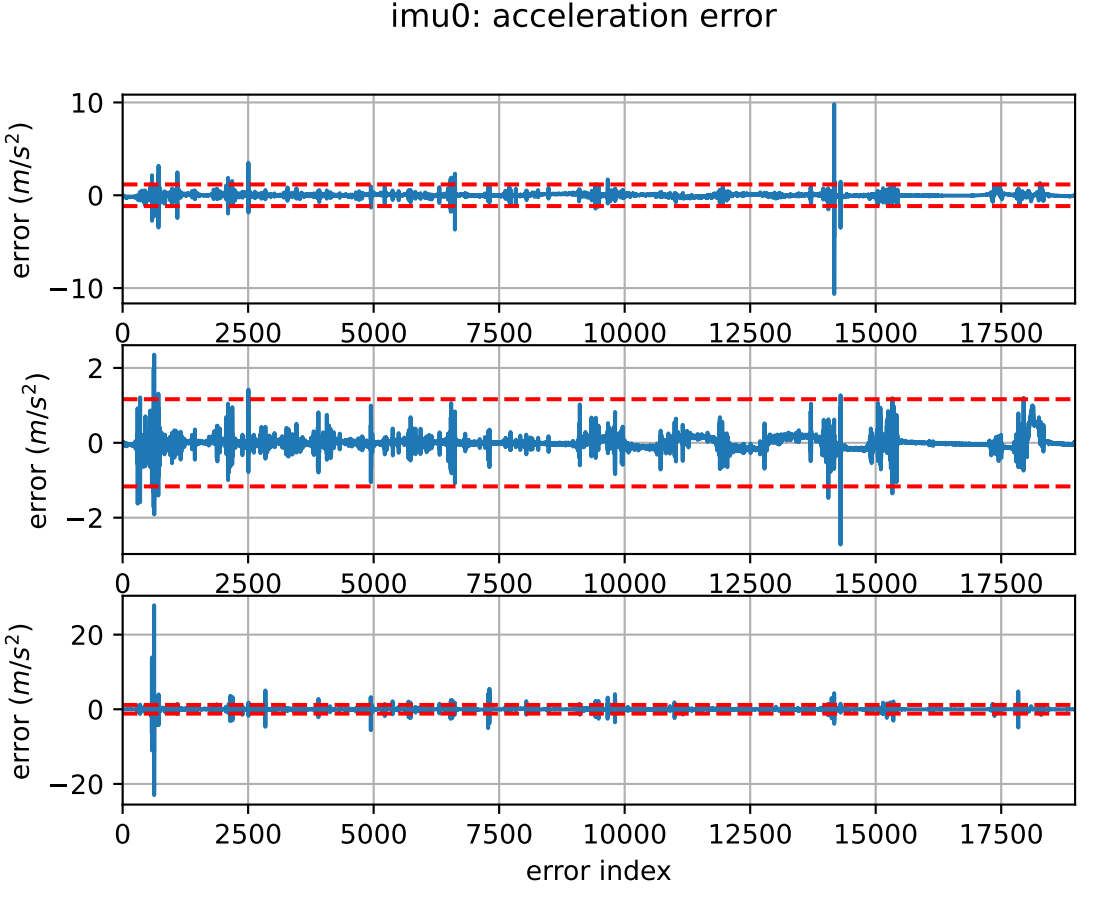
  Model: calibrated
  Update rate: 204.090470950253
  Accelerometer:
    Noise density: 0.027176766288825538
    Noise density (discrete): 0.38824792519257595
    Random walk: 0.0067023621585376265
  Gyroscope:
    Noise density: 0.002763830369432253
    Noise density (discrete): 0.039484145947029654
    Random walk: 0.00014770260860981916

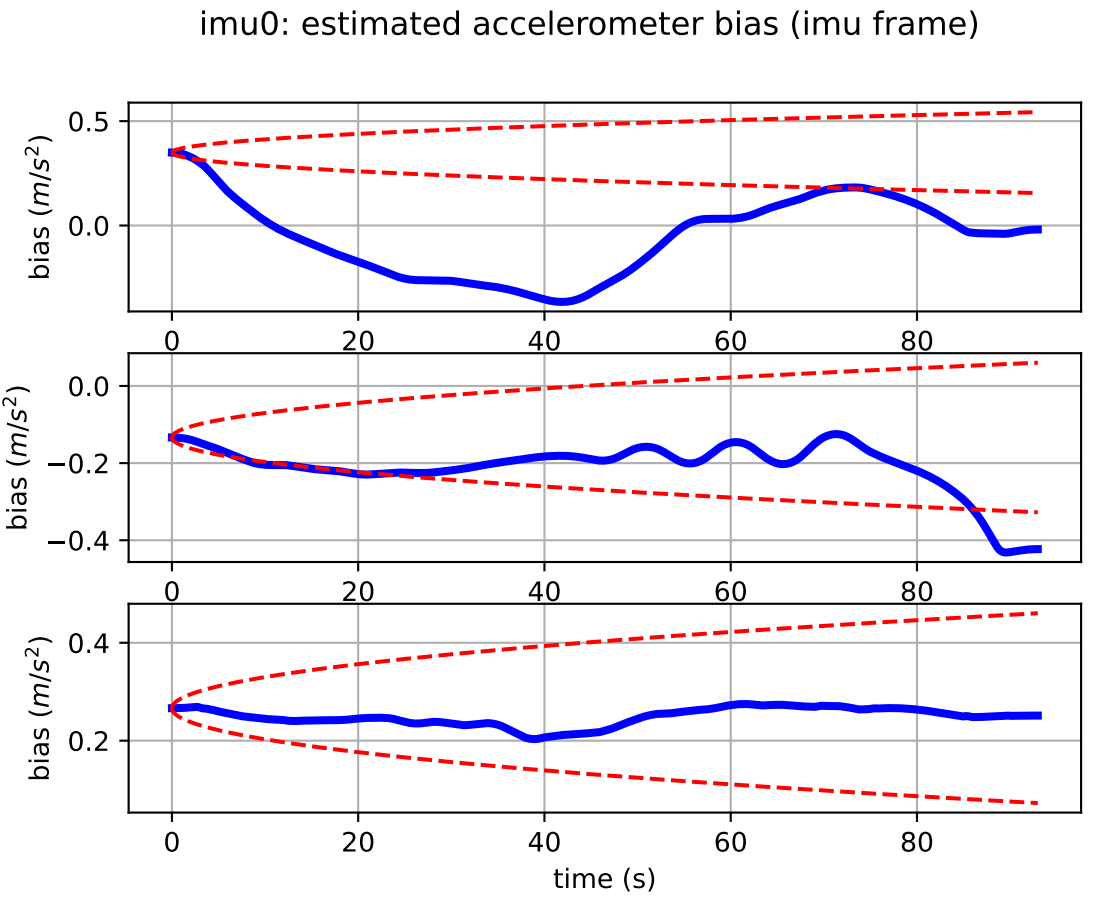
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

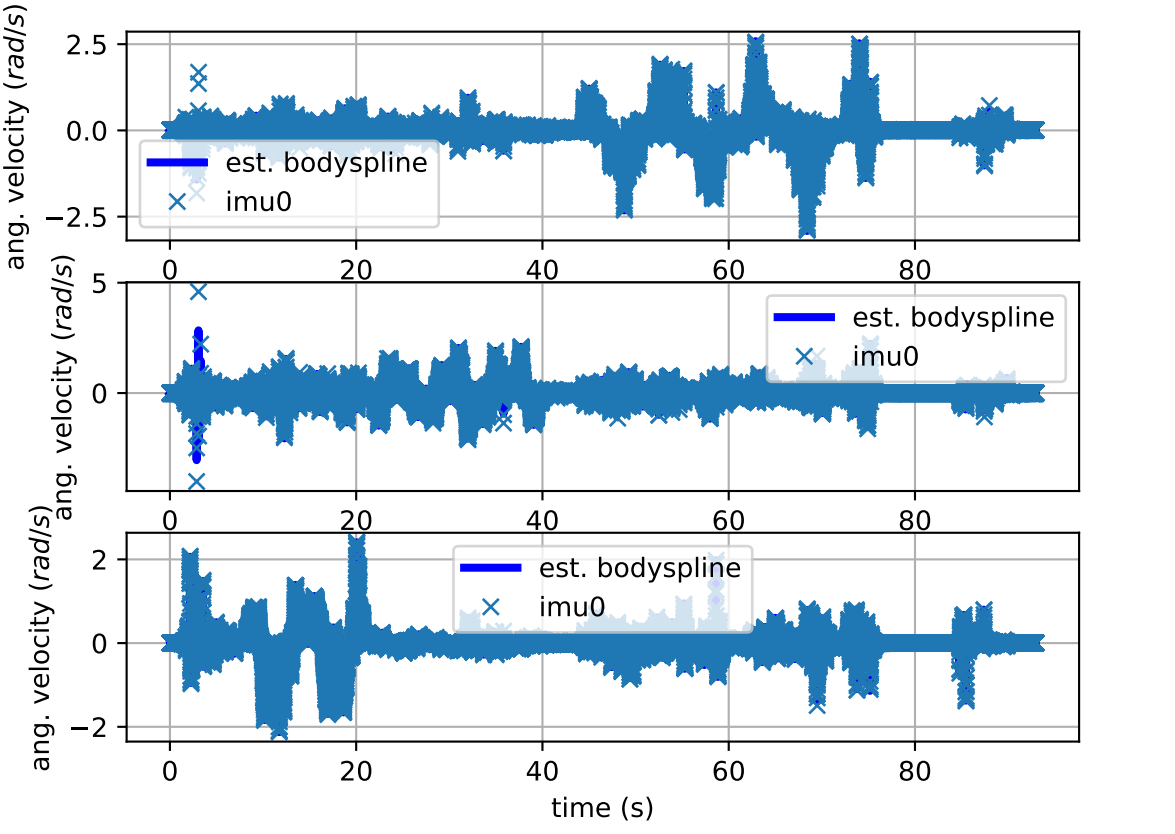

imu0: estimated poses
0.50
0.45
0.40
0.35
0.30
0.05
0.00
−0.05
−0.10
−0.15
−0.20
0.00
0.05
0.10
0.15
0.20

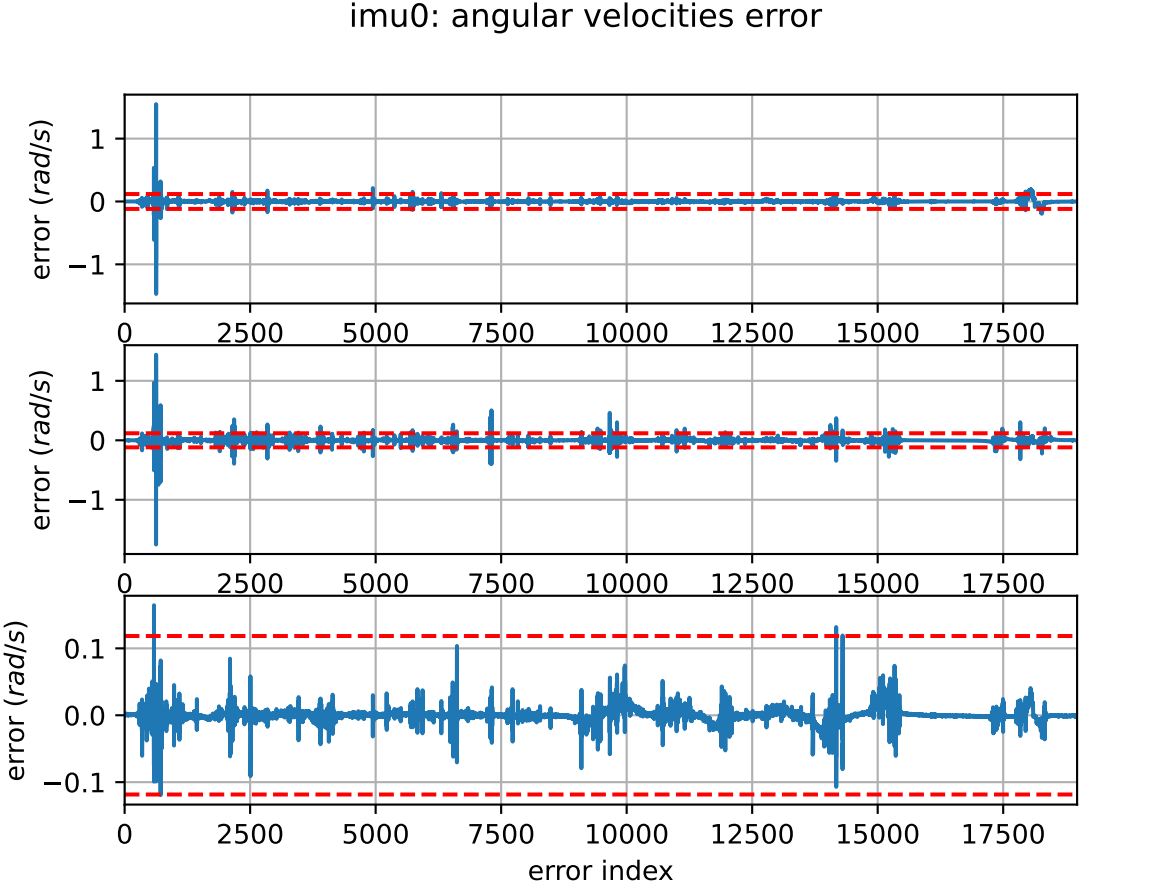

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0000
sample rate (ms)
0
1
2
3
4
20
40
60
80
time (s)

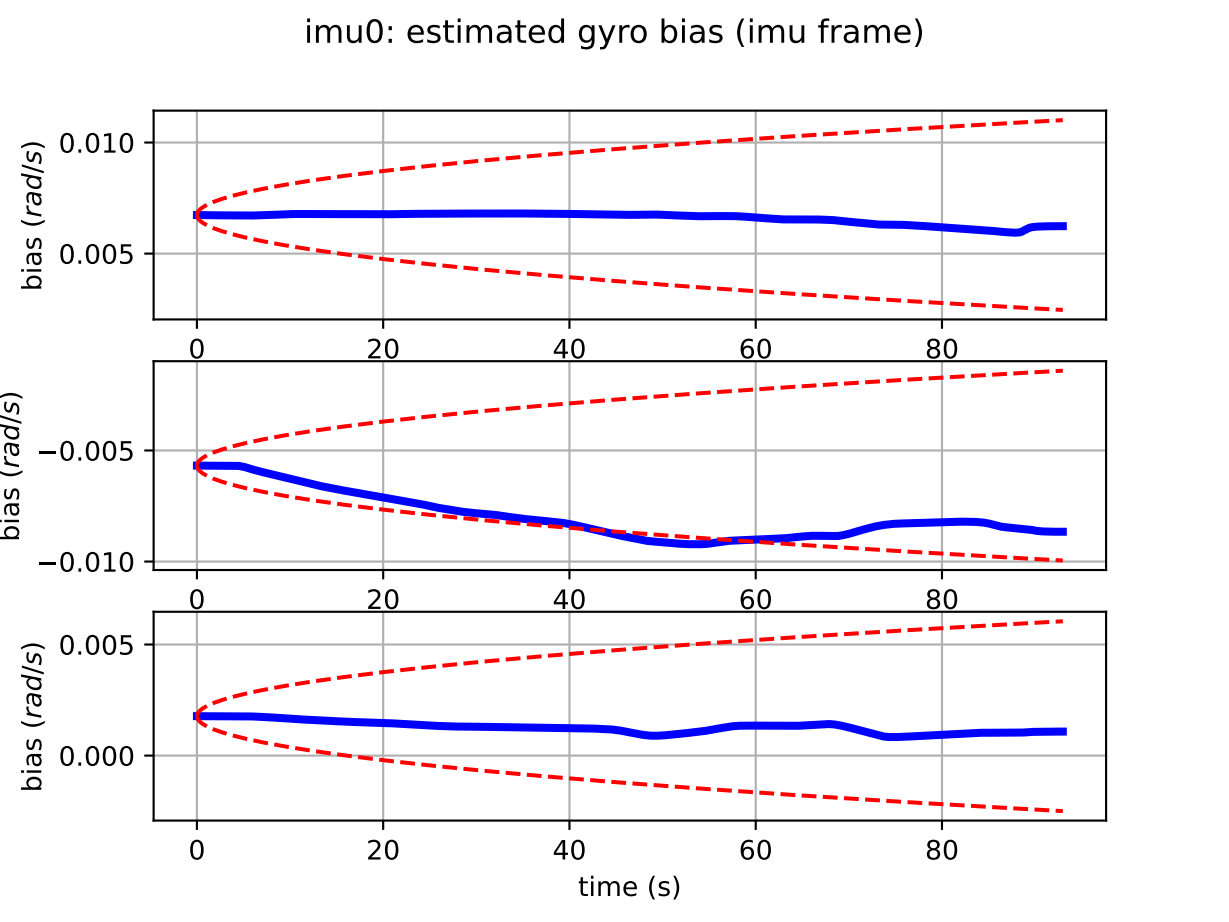

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
10
0
−10
0
−10
20
0
−20
0
20
40
60
80
time (s)

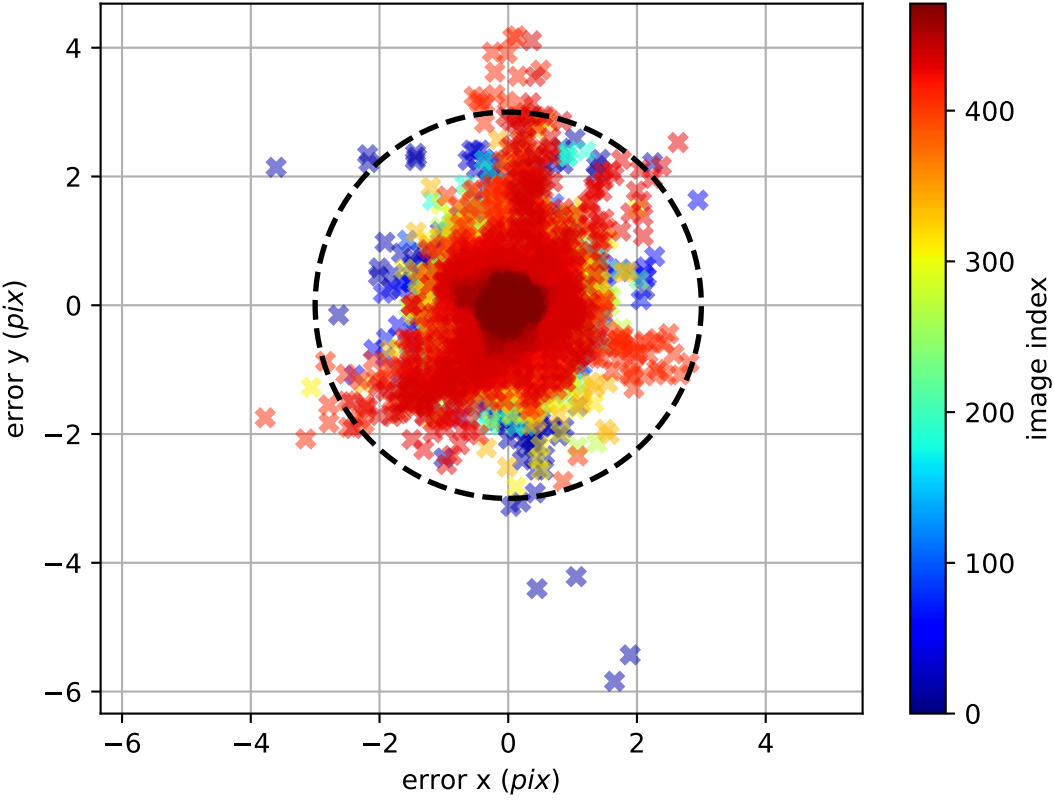

imu0: acceleration error
error (m/s^2)
10
0
−10
0
2500
5000
7500
10000
12500
15000
17500
error (m/s^2)
2
0
−2
0
2500
5000
7500
10000
12500
15000
17500
error (m/s^2)
20
0
−20
0
2500
5000
7500
10000
12500
15000
17500
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.5
0.0
0
20
40
60
80
bias (m/s^2)
0.0
−0.2
−0.4
0
20
40
60
80
bias (m/s^2)
0.4
0.2
0
20
40
60
80
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
2.5
0.0
−2.5
0
20
40
60
80
5
0
2
−2
time (s)


imu0: angular velocities error
error (rad/s)
1
0
−1
0
2500
5000
7500
10000
12500
15000
17500
error (rad/s)
1
0
−1
0
2500
5000
7500
10000
12500
15000
17500
error (rad/s)
0.1
0.0
−0.1
0
2500
5000
7500
10000
12500
15000
17500
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.010
0.005
0
20
40
60
80
bias (rad/s)
−0.005
−0.010
0
20
40
60
80
bias (rad/s)
0.005
0.000
0
20
40
60
80
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−6
−4
−2
0
2
4
0
100
200
300
400